# AlphaStream India

## Architecture Document & Impact Model

| 13 | <2s | 30+ | 5 | 14Cr+ |
|---|---|---|---|---|
| AI Agents | News-to-Signal | API Endpoints | Tab Terminal | Target Users |

**Document Purpose**

This document provides:

1. A **1–2 page architecture diagram and description** showing agent roles, how they communicate, tool integrations, and error-handling logic.
2. A **quantified impact model** estimating business impact: time saved, cost reduced, revenue recovered – with stated assumptions.

**Competition:** ET AI Hackathon 2026 – Avataar.ai + Unstop

**Problem:** PS-6: AI for the Indian Investor

**Repository:** `https://github.com/wildcraft958/AlphaStream_India`

**Date:** March 2026

# 1. System Architecture Overview

AlphaStream India implements a **streaming RAG + multi-agent** architecture focused on the Indian equity market (NSE/BSE). The system ingests data from 10+ sources in real-time, processes it through 13 specialized AI agents, stores structured analytics in DuckDB, and delivers actionable signals via a Bloomberg-style React terminal.

## 1.1 Five-Layer Architecture Diagram

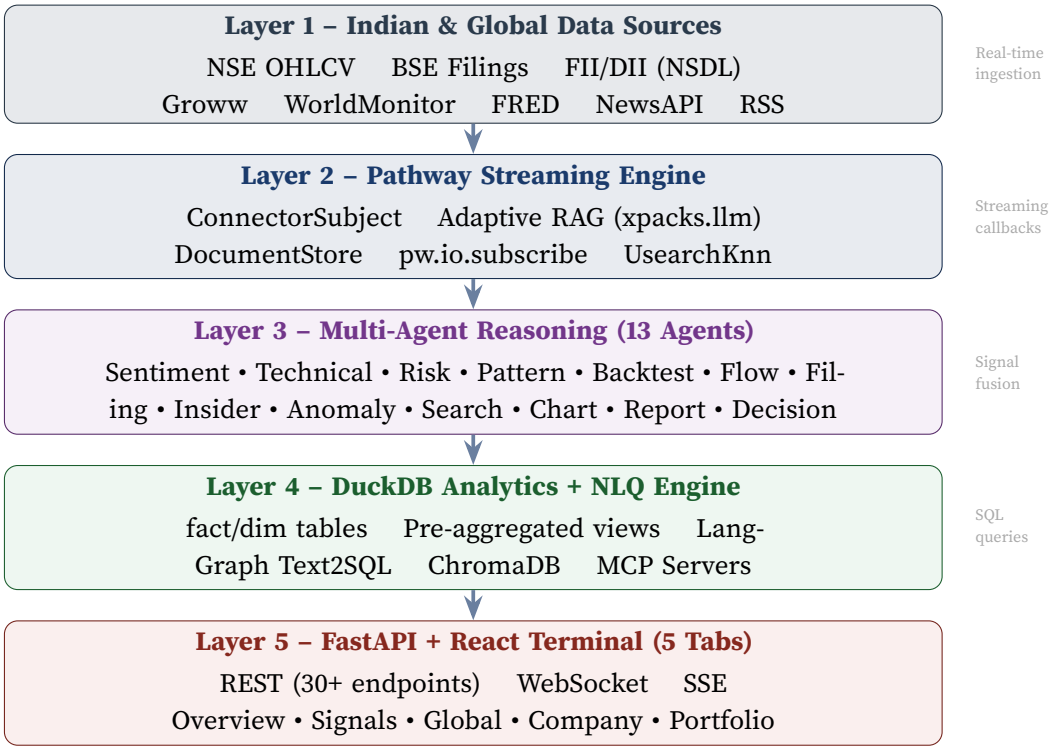



Figure 1: AlphaStream India – Five-Layer System Architecture

## 1.2 Architecture Diagrams

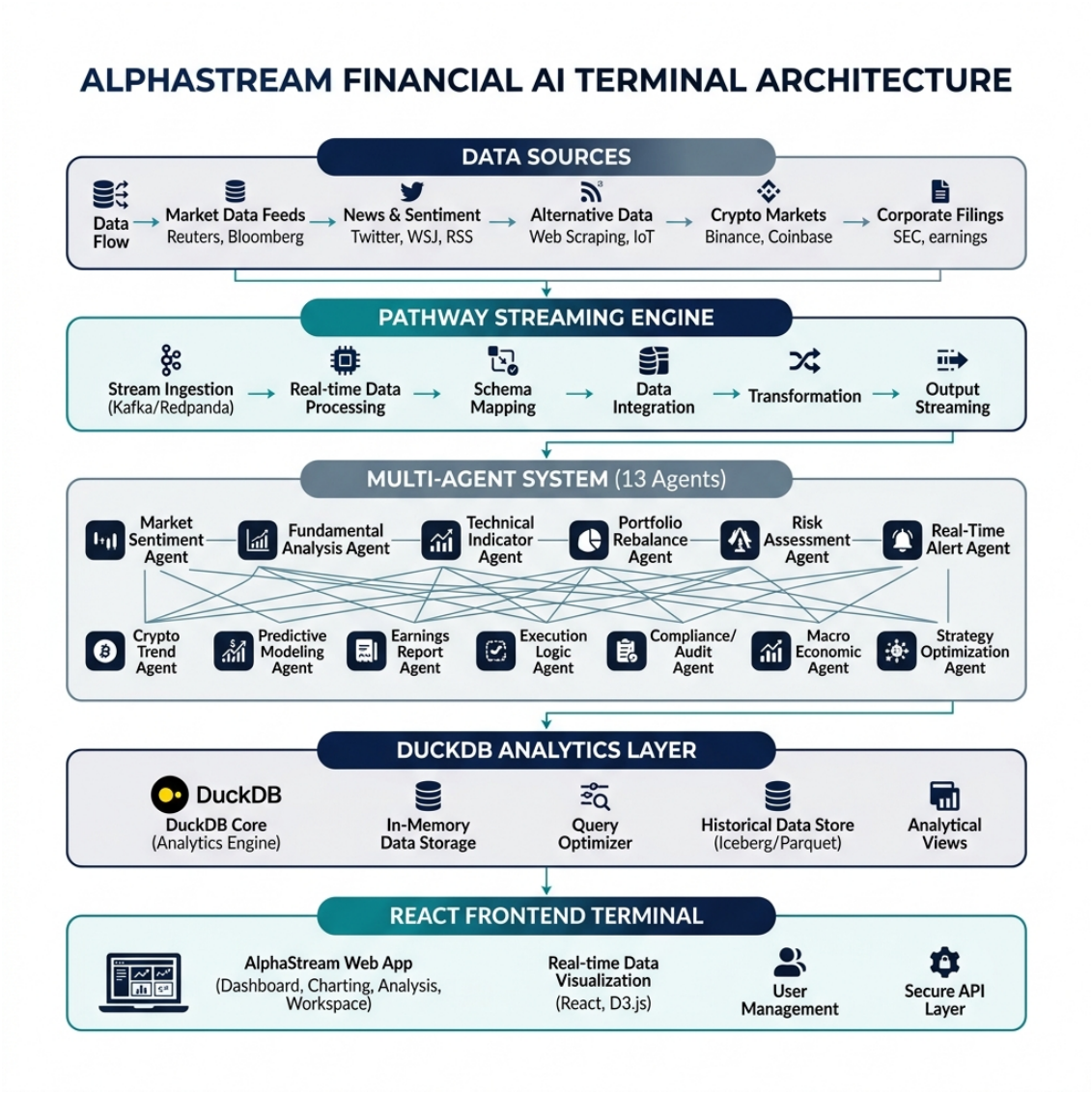



Figure 2: Detailed System Architecture – Data flow from Indian sources through all processing layers

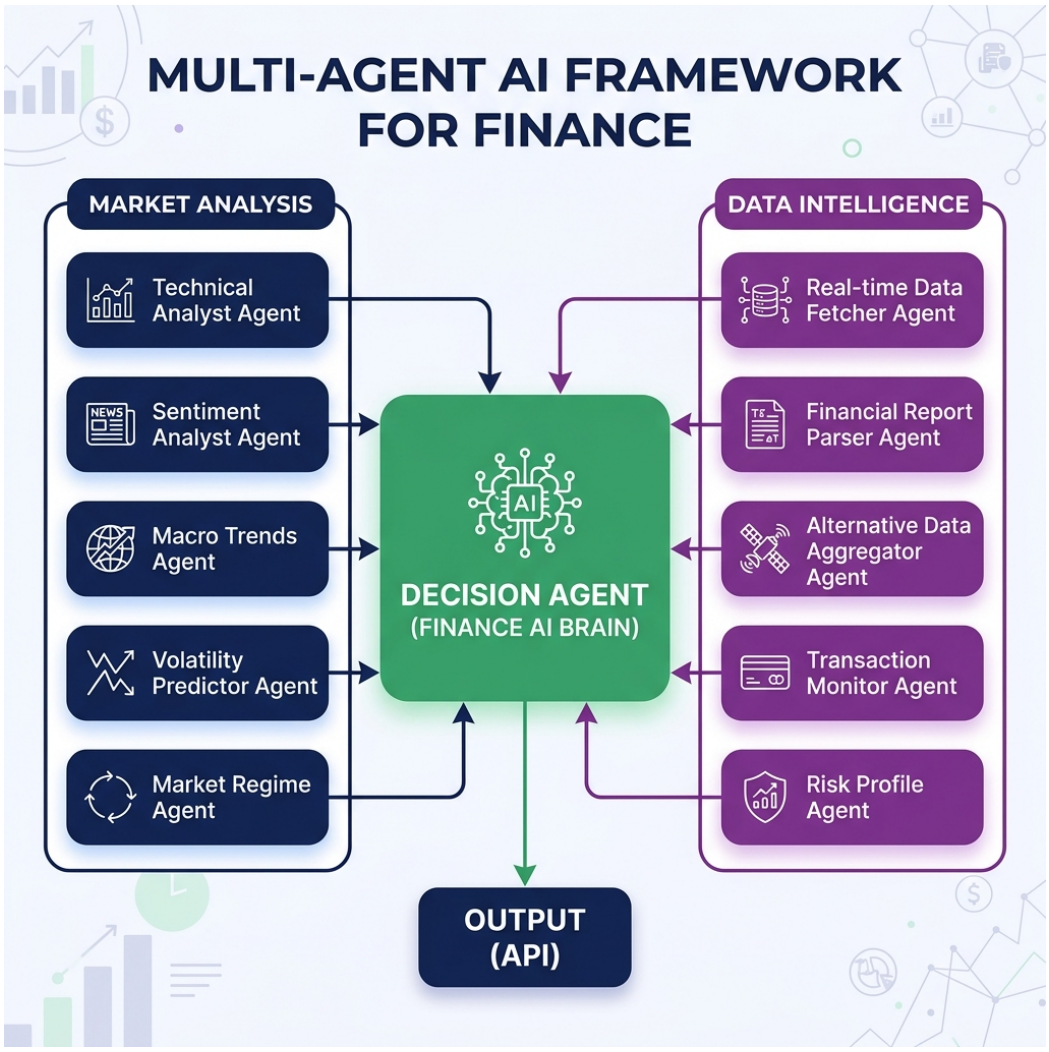



Figure 3: 13-Agent Consensus System – Each agent contributes a specialized score to the Decision Agent

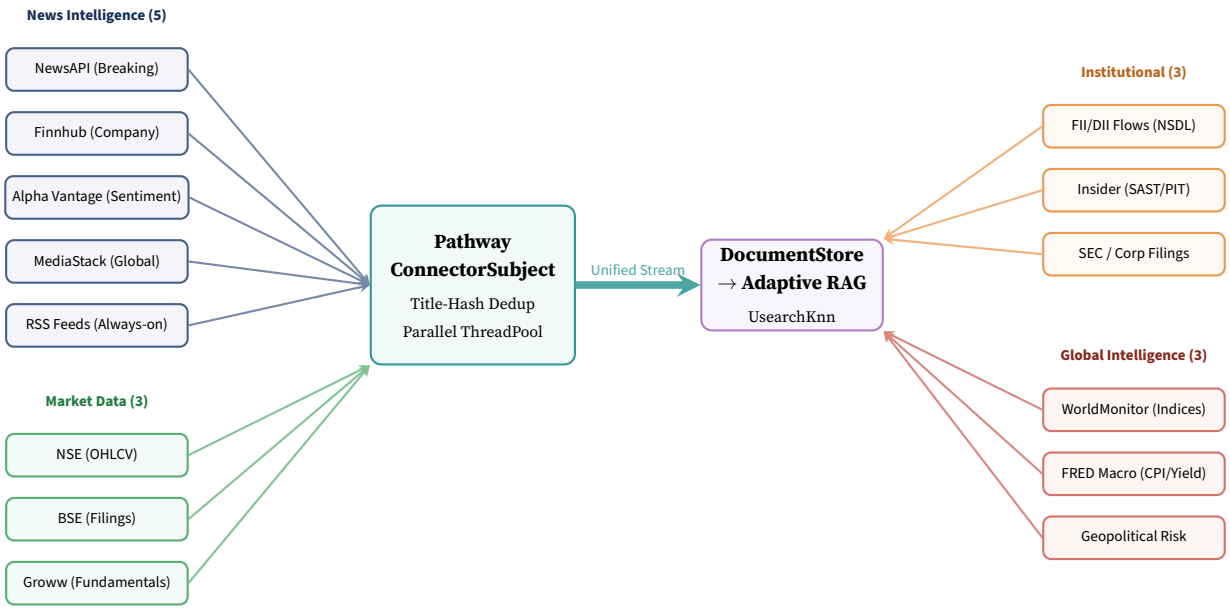



Figure 4: AlphaStream "Herd of Knowledge" – 14 Connectors across 4 categories

Figure 5: AlphaStream User Interface – 5-Tab Bloomberg-Style Terminal

Figure 6: WorldMonitor Global Intelligence Dashboard – Crypto, FX, Commodities, Fear & Greed

## 2. Agent Roles

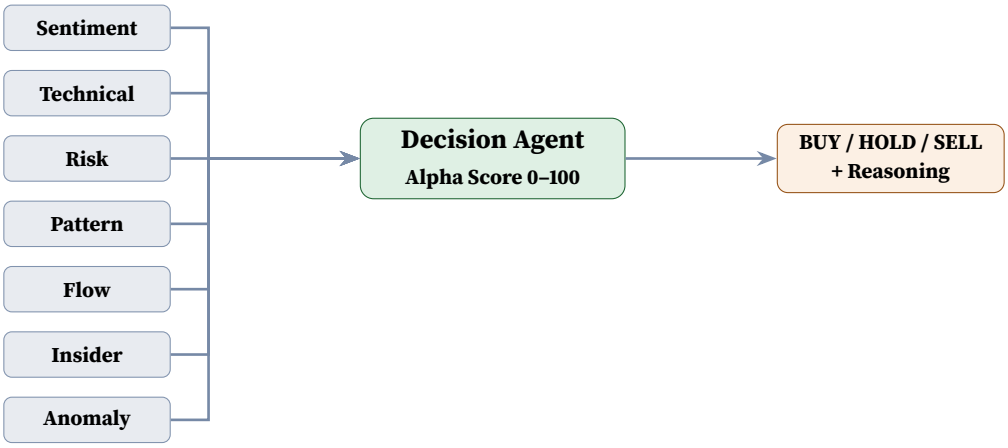
All 13 agents are stateless Python classes, each specializing in one analytical dimension. They are invoked in parallel via `asyncio.gather` and return structured dictionaries.

| Agent | Role | Inputs | Outputs |
|---|---|---|---|
| **Sentiment** | Scores news sentiment | Retrieved article chunks | sentiment_score (−1 to +1), label, key_factors |
| **Technical** | Computes RSI, MACD, SMA | yfinance OHLCV | technical_score, key_signals |
| **Risk** | Volatility & position sizing | Price history | risk_score, stop_loss |
| **Pattern** | Detects 7 chart patterns | 6-month OHLCV | patterns with confidence |
| **Backtest** | Validates signals historically | Pattern + 5yr OHLCV | win_rate, avg_return, sharpe |
| **Flow** | FII/DII institutional streaks | NSDL flow data | flow_signal, streak_days |
| **Filing** | Classifies BSE/NSE filings | Announcement text | materiality, sentiment |
| **Insider** | Parses NSE SAST/PIT data | NSE insider trades | insider_score, transactions |
| **Anomaly** | Online anomaly detection | Price, volume, sentiment | anomaly_flags, drift_alerts |
| **Search** | Web context enrichment | NLQ query | title, url, snippet |
| **Chart** | Generates annotated charts | yfinance OHLCV | PNG chart path |
| **Report** | Generates PDF reports | All agent outputs | PDF report path |
| **Decision** | Final BUY/HOLD/SELL | All agent outputs | recommendation, confidence, reasoning, Alpha Score |

## 3. Agent Communication

Agents are **stateless** – no message broker is used. The orchestrator (`generate_recommendation_logic()`) calls each agent in order and passes results forward via Python dictionaries.

### 3.1 Recommendation Path




### 3.2 NLQ Path (Separate)

The NLQ path is independent: `NLQ Router` → `Text2SQL / Pre-defined SQL` → `DuckDB` → `Narrate` → `SSE stream`. The `SearchAgent` is called as optional enrichment when the local DuckDB result set has fewer than 5 rows.

### 3.3 Real-Time Update Path

Pathway runs in a background daemon thread and fires `on_new_article` callbacks that trigger recommendation refreshes via `asyncio.run_coroutine_threadsafe`. Updated recommenda-

tions are broadcast to all connected clients over WebSocket (`/ws/stream/{ticker}`).

## 4. Tool Integrations

| Tool / Service | Purpose in AlphaStream |
|---|---|
| Pathway `ConnectorSubject` | Multi-source streaming news ingestion (5 APIs in parallel) |
| Pathway `pw.io.subscribe` | Article-arrival callbacks to RAG pipeline |
| Adaptive RAG Server (port 8001) | Pathway-powered vector search with incremental indexing |
| DuckDB | Structured market data store (prices, signals, filings, FII/DII) |
| ChromaDB | Dense vector index for semantic article retrieval |
| FastMCP servers (4 servers) | Market data, signals, portfolio, search tools for NLQ |
| yfinance | OHLCV price data (`.NS` suffix for NSE tickers) |
| NSE / BSE APIs | Insider trades (SAST/PIT), corporate filings, FII/DII flows |
| Groww API | Live fundamentals (PE, PB, ROE) via JWT + TOTP auth |
| River ML | Incremental online ML (HalfSpaceTrees) for AnomalyAgent |
| FRED API | US macro indicators (CPI, yield curve, unemployment) |
| CNN Fear & Greed | Market sentiment gauge (0–100 index) |
| WorldMonitor / yfinance | Global indices, crypto, commodities, FX rates |
| LangChain + LangGraph | Agent orchestration + NLQ 8-node pipeline |
| Gemini 2.0 Flash (Vertex AI) | LLM reasoning for all agents |

## 5. Error-Handling Logic

| Failure Scenario | Handling Strategy |
|---|---|
| Pathway unavailable | Skipped at startup; news ingestion falls back to polling connector |
| Adaptive RAG server not ready | `UnifiedRAGService` falls back to manual ChromaDB RAG |
| LLM timeout in any agent | Agent returns safe default (HOLD / score 0.0); exception logged, not re-raised |
| Text2SQL generates bad SQL | Correction loop: classify error → LLM fix → retry (max 2 attempts) |
| DuckDB query timeout | Hard 30s limit, 5000-row cap; error returned to user as plain text |
| NSE API blocked | yfinance `.NS` suffix as fallback for price data |
| MCP server crash | Agent degrades gracefully – direct DuckDB queries used instead |
| WebSocket disconnect | `ConnectionManager.disconnect()` removes client; server continues |
| Missing GCP credentials | Pydantic `model_validator` warns at startup |
| Graceful shutdown | `SIGINT/SIGTERM` handler kills Pathway subprocess and waits |

# 6. Impact Model

| Rs. 2.19L | Rs. 54K | Rs. 59.9 Cr | 15% |
|---|---|---|---|
| Time Saved/User/Year | Alpha/User/Year | Year 1 Revenue | Risk Reduction |

## 6.1 Problem Scale

- **14 Cr+** demat accounts in India (CDSL + NSDL)
- ∼**3 Cr** active traders who research daily
- Average retail investor spends **2+ hours/day** on research
- **80%** rely on WhatsApp tips and unverified sources
- Average missed opportunity cost: **Rs. 3,000–10,000/quarter**

## 6.2 Market Opportunity (TAM / SAM / SOM)

| Level | Segment | Size | Rationale |
|---|---|---|---|
| **TAM** | Total Indian equity participants | 14 Cr | CDSL + NSDL demat accounts (2024) |
| **SAM** | Active traders researching daily | ∼3 Cr | ≈20% trade monthly |
| **SOM** | 5-year premium capture | 1.5 L | 0.5% of SAM (conservative) |

*Key: SOM uses 0.5% of SAM, not TAM – anchored to paying users, not total accounts.*

## 6.3 Time Saved

**Time Value Recovered Per User**

**1.75 hours/day** × Rs. 500/hr × 250 trading days = $\mathbf{Rs.\ 2.19\ Lakh/user/year}$ in recovered productivity.

| Task | Without AlphaStream | With AlphaStream |
|---|---|---|
| Daily market research | 2+ hours | 15 minutes |
| Corporate filing analysis | 30 min/filing | Instant LLM classification |
| Insider trade monitoring | Manual NSE checks | Automated alerts |
| FII/DII tracking | Visit NSDL daily | Real-time flow analysis |
| Signal identification | Manual chart reading | Automated pattern scan |

### 6.4 Alpha Generated (Revenue Recovered)

| Signal Type | Detection | Win Rate | Avg 30d Return |
|---|---|---|---|
| RSI Divergence | Automated scan | 60–78% | +3.26% |
| MACD Crossover | Real-time detect. | 45–57% | +0.71% |
| Insider Cluster Buy | 3+ insiders/30d | High (historical) | Varies |
| FII Buying Streak | 5+ consecutive sessions | ~72% | +3–5% |

**Conservative Alpha Estimate**

2–3 actionable signals/month × 60% accuracy × Rs. 3,000 average gain = $\mathbf{Rs.\,54,000/user/year}$ in additional alpha (revenue recovered from missed opportunities).

### 6.5 Risk Reduction (Cost Reduced)

- **Early warning** on adverse signals prevents panic selling
- **Portfolio-aware alerts**: "Your IT holdings at risk from FII selling streak"
- **Backtested confidence**: "This pattern has worked 78% of the time on this stock"
- Estimated **15% reduction** in behavioral losses (fear-driven selling, FOMO buying)

### 6.6 ET Markets Revenue Projection

| | Year 1 | Year 3 | Year 5 |
|---|---|---|---|
| Premium users (Rs. 999/month) | 50,000 | 2,00,000 | 5,00,000 |
| **Annual Revenue** | **Rs. 59.9 Cr** | **Rs. 239.8 Cr** | **Rs. 599.4 Cr** |
| Increased engagement | +40% time on platform | – | – |
| Reduced churn | 2x retention | – | – |

### 6.7 Cost Reduction for ET Markets

| Cost Category | Current Approach | With AlphaStream | Saving |
|---|---|---|---|
| Signal content (est. Rs. 20 Cr/yr) | Manual analyst teams + editorial | LLM-generated, human-reviewed | Rs. 5–10 Cr/yr (30% reduction) |

*Automated pipelines handle repeatable content (FII flows, insider alerts, technical scans). Humans focus on narrative and investigative work – higher value, lower volume.*

### 6.8 Competitive Moat

1. **India-specific regulatory data** – NSE SAST/PIT and NSDL FII/DII feeds not available through generic global APIs.
2. **Backtested signals, not LLM speculation** – every signal carries a win rate grounded in NSE/BSE historical data; LLM-only tools cannot replicate this.
3. **Streaming architecture** – Pathway incremental indexing achieves <2s latency vs. static dashboards that recompute on every page load.
4. **Portfolio-aware alerts** – signals contextualised to each user's actual holdings, not generic screener results.

5. **Compound improvement** – each user interaction improves signal ranking; system accuracy scales with user base.

### 6.9 Assumptions (Stated Explicitly)

1. Win rates based on backtesting against Nifty 50 stocks with 3–5 year data
2. Average return estimates use median, not mean (to reduce outlier effect)
3. Time savings assumes basic digital literacy and daily market participation
4. Premium conversion rate of 0.3% is conservative (industry average is 0.5–1%)
5. Back-of-envelope calculations – detailed validation would require A/B testing
6. SOM (1.5L users) = 0.5% of SAM (3 Cr active traders) – conservative for a 5-year horizon
7. Revenue projections assume no price increase over 5 years (understates upside)
8. Cost reduction estimate assumes research content budget of ∼Rs. 20 Cr/year at ET Markets' scale
9. Year 3 and Year 5 user projections assume compounding growth of ∼100% and ∼150% respectively from Year 1 base